

Hydrogen Production from Biogas by Dry Reforming and Water-Gas Shift: An Aspen Plus Simulation Study

A Technical Simulation Report

Prepared Using
Aspen PLUS V14

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Abstract

Biogas — the mixture of methane and carbon dioxide produced by anaerobic digestion of organic waste — is one of the more underused feedstocks in the current hydrogen landscape. This study uses Aspen Plus V14.0 to simulate a continuous hydrogen production process from biogas via dry reforming (DRM) followed by a dual-stage water-gas shift (WGS) reaction and pressure swing adsorption (PSA) purification. The biogas feed, at 38.51 kg/h and a composition of 59.7 vol% CH₄, 40.06 vol% CO₂, 0.2 vol% N₂, and 0.04 vol% O₂, enters a Gibbs free energy minimization reformer operating at 909 °C and 16 bar. Syngas from the reformer passes through a heat-integrated two-stage WGS section — high-temperature at 350 °C and low-temperature at 210 °C — each achieving 75% CO conversion before the stream reaches a PSA unit at 38 °C and 15.65 bar with a hydrogen separation efficiency of 79%. The tail gas from PSA is combusted with air to supply heat to the system. The simulation converged in three Wegstein iterations with a maximum relative error of 9.1×10^{-11} . Net CO₂-equivalent reduction across the process is 292.26 kg/hr, which reflects the carbon utilization advantage of treating CO₂-rich biogas as a reactant rather than a waste stream.

1. Introduction

The global push for clean hydrogen has renewed interest in routes that do not start with fossil natural gas. Biogas — produced from landfills, wastewater sludge, agricultural residues, and municipal organic waste — offers an alternative feedstock that is both renewable and already carbon-loaded. Roughly 40 to 50 percent of typical biogas is CO₂, which conventional hydrogen production via steam methane reforming (SMR) would treat as an inert contaminant. Dry reforming uses that CO₂ as a co-reactant, converting both CH₄ and CO₂ into syngas simultaneously.

The chemistry is attractive. The economics are harder. Dry reforming runs at temperatures well above 800 °C, requires substantial heat input, and produces significant coke under certain operating regimes. Coupling DRM with WGS addresses the downstream H₂:CO ratio problem — raw DRM syngas sits near H₂:CO = 1, while hydrogen-optimized processes need ratios closer to 3:1 or higher depending on the separation method. The WGS reaction converts CO and steam to H₂ and CO₂, raising hydrogen content at the cost of some CO₂ generation. A PSA unit then isolates high-purity hydrogen from the remaining gas mixture.

This simulation assembles those unit operations into a heat-integrated flowsheet and uses Aspen Plus equilibrium and stoichiometric reactor models to quantify stream compositions, energy duties, and the overall hydrogen yield from a defined biogas input. The property method used throughout is PR-MHV2, a modified Peng-Robinson equation of state that handles the high-temperature, high-pressure, and mixed polar-nonpolar component system present in this process. No external utility streams are assumed for the reformer — heat is supplied internally by combustion of PSA tail gas with air.

The objectives are straightforward: build a converged steady-state simulation of the integrated DRM-WGS-PSA process, extract key performance metrics (hydrogen production, CO conversion, energy duties, CO₂ balance), and produce a documented basis for further optimization or techno-economic analysis.

2. Process Description

The flowsheet consists of five functional sections: feed compression and pre-heating, dry reforming, syngas cooling and mixing, two-stage water-gas shift, and hydrogen purification with tail gas combustion. Figure 1 shows the Aspen Plus process flow diagram as simulated.

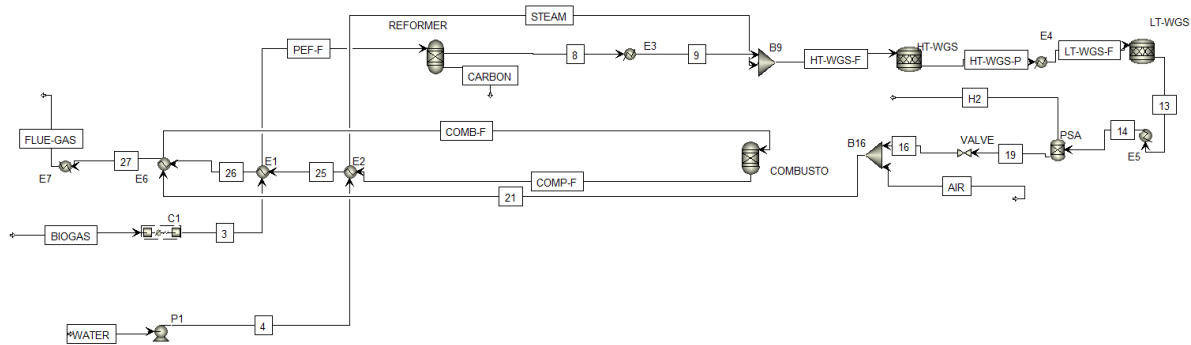


Figure 1: Aspen Plus PFD — Hydrogen Production from Biogas via DRM + WGS

2.1 Feed Section

Biogas enters the system at ambient conditions (25 °C, 1 bar) with a mass flow of 38.51 kg/h. Before it can enter the reformer, which operates at 16 bar, the gas must be compressed. Compressor C1 is a three-stage isentropic centrifugal compressor with interstage cooling. Each stage operates at a pressure ratio of 2.52, and the first two stages are cooled to 60 °C after compression. The third stage delivers the gas at 16 bar and approximately 155 °C. Total compressor work is 4.26 kW, with intercooling duties of 198.2 and 360.1 cal/s in stages 1 and 2 respectively.

The compressed biogas stream (stream 3) then passes through heat exchanger E1, where it is preheated against the hot product gas leaving E1's hot side (stream 25, combustion flue at 2128 °C). This brings the biogas to 909 °C before it enters the reformer as stream PEF-F. The countercurrent arrangement in E1 achieves a log-mean temperature difference of 1381.8 °C, with a heat duty of 4111 cal/s across a calculated area of 0.0147 m².

Water enters as a separate feed at 25 °C and 1 bar (8.70 kg/h, 0.483 kmol/hr). Pump P1 raises pressure to 16 bar with a brake power of 0.0057 kW and 75% efficiency. The pressurized water (stream 4) is then fully vaporized and superheated in heat exchanger E2, using combustion exhaust at 2302 °C as the heat source. The outlet steam (stream STEAM) reaches 350 °C at 16 bar, with a boiling point at this pressure of approximately 201.5 °C. Steam generation duty in E2 is 1757 cal/s.

2.2 Dry Reforming Reactor

The reformer is modeled as an RGibbs reactor — a Gibbs free energy minimization block that finds the equilibrium product distribution for the given inlet composition, temperature, and pressure without requiring a pre-specified reaction stoichiometry. This is appropriate for dry reforming, where multiple reactions occur simultaneously and the equilibrium product distribution is sensitive to temperature and pressure. The reactor operates at 909 °C and 16 bar. Carbon graphite (solid C) is included in the product list to allow coke formation, which the equilibrium calculation predicts at 0.412 kmol/hr.

The feed (PEF-F) enters at 909 °C, 16 bar, carrying 0.843 kmol/hr CH₄ and 0.566 kmol/hr CO₂ along with trace N₂ and O₂. The reformer output (stream 8) contains 0.169 kmol/hr CH₄, 0.100 kmol/hr CO₂, 0.728 kmol/hr CO, 1.144 kmol/hr H₂, 0.204 kmol/hr H₂O, and 0.412 kmol/hr solid

carbon. The endothermic heat duty on the reformer is 7534 cal/s, which in the simulation is supplied by the combustor side of the flowsheet.

A CH₄ conversion specification of 20% remaining (i.e., 80% conversion) was set in the RGibbs block to constrain the equilibrium toward realistic operating conditions. The resulting syngas has an H₂:CO molar ratio of approximately 1.57, lower than SMR-derived syngas (typically 3:1) and consistent with the expected DRM product distribution.

2.3 Syngas Cooling and Mixing

Hot syngas at 909 °C exits the reformer as stream 8 and passes through heater block E3, which cools it to 350 °C at 16 bar. In the simulation, E3 is modeled as a simple heater (TP flash), removing 3085 cal/s. This cooling step serves two purposes: it protects downstream equipment from thermal stress and brings the syngas to the inlet temperature required for the high-temperature WGS reactor.

Cooled syngas (stream 9) then mixes with the superheated steam (stream STEAM) in mixer B9. The combined stream HT-WGS-F has a total flow of 2.831 kmol/hr at 350.2 °C and 16 bar, with a composition dominated by H₂ (40.5 mol%), CO (25.7 mol%), H₂O (24.3 mol%), and CO₂ (3.5 mol%). Adding steam at this stage raises the H₂O:CO ratio, which is the thermodynamic driver for the subsequent WGS reactions.

2.4 Water-Gas Shift Section

The WGS section uses two sequential RStoic reactors with the same reaction stoichiometry ($\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$) but different operating conditions, matching the industrial dual-stage configuration that uses iron-chromium oxide catalyst at high temperature and copper-zinc oxide catalyst at low temperature.

The high-temperature WGS reactor (HT-WGS) operates at 16 bar with a CO conversion of 75%. Starting from 0.728 kmol/hr CO in HT-WGS-F, this converts 0.546 kmol/hr CO and produces an equivalent amount of H₂, raising the H₂ content of the exit stream (HT-WGS-P) to 1.690 kmol/hr. The adiabatic outlet temperature calculated is 551.2 °C, consistent with the exothermic nature of WGS. The heat duty at zero (PQ flash with zero duty specified) means the temperature rise is a simulation output, not an input constraint.

Before entering the low-temperature stage, stream HT-WGS-P is cooled from 551.2 °C to 210 °C by heater E4, removing 2304 cal/s. The LT-WGS reactor operates at 15.7 bar with the same 75% CO conversion specification applied to the remaining CO. The inlet CO to LT-WGS is 0.182 kmol/hr; 75% conversion produces an additional 0.137 kmol/hr of H₂. The LT-WGS outlet temperature is 264 °C, again reflecting the exothermic heat release. Stream 13 exits LT-WGS with 1.827 kmol/hr H₂, 0.782 kmol/hr CO₂, 0.169 kmol/hr CH₄, and 0.045 kmol/hr residual CO.

2.5 Purification by Pressure Swing Adsorption

Stream 13 passes through heater E5, which cools it to 38 °C at 15.7 bar before entering the PSA unit. Cooling removes 1446 cal/s. The PSA is modeled as a Sep block with defined split fractions. All impurities (CH₄, CO₂, CO, H₂O, N₂) are directed entirely to the tail gas stream (stream 19), while 21% of the H₂ reports to stream 19 and 79% is recovered as product H₂.

This split is intentional: in the simulation context, the PSA efficiency of 79% refers to the fraction of hydrogen sent to the product stream H₂ (stream H2). The tail gas stream 19 carries 1.443 kmol/hr H₂ along with all impurity components. It passes through valve VALVE, which drops pressure from 15.7 bar to 1 bar (a pressure drop of 14.7 bar) before entering mixer B16 where it combines with the air stream for combustion.

2.6 Combustion and Heat Integration

The PSA tail gas, after pressure reduction, mixes with an air stream (79.68 kg/h, 21 vol% O₂ / 79 vol% N₂) in mixer B16. The combined stream (21) enters heat exchanger E6, where it is pre-heated from 29.4 °C to 250 °C using hot gas from stream 26 (flue gas train at 1713 °C). E6 transfers 1811 cal/s. The pre-heated combustion mixture (COMB-F) then enters the combustor (COMBUSTO), modeled as an RGibbs block at 1 bar with zero heat duty — adiabatic equilibrium combustion. The calculated combustion temperature is 2302.6 °C.

Hot combustion products (COMP-F) at 2303 °C flow back through E2 (steam generation, 1757 cal/s removed) and E1 (biogas pre-heating, 4111 cal/s removed), cooling progressively from 2303 °C to 2128 °C to 1713 °C in sequence. Remaining hot gas passes through E6 again (stream 26 to 27, 1713 °C to 1526 °C), then through final cooler E7 where it is brought to 200 °C and exits as FLUE-GAS. E7 removes 11484 cal/s, which represents the net heat rejected to the environment.

3. Reaction Chemistry

The process involves three main reaction systems operating at different temperatures and conditions. Table 1 summarizes the key reactions.

No.	Reaction	Type	ΔH° (kJ/mol)	T (°C)
1	$\text{CH}_4 + \text{CO}_2 \rightarrow 2\text{CO} + 2\text{H}_2$	DRM	+247	800–950
2	$\text{CH}_4 \rightarrow \text{C} + 2\text{H}_2$	Methane cracking	+75	>600
3	$2\text{CO} \rightarrow \text{C} + \text{CO}_2$	Boudouard	−172	<700
4	$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	HT-WGS	−41	310–450
5	$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	LT-WGS	−41	200–250
6	$\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$	Combustion	−242	~2300

Table 1: Key reactions in the DRM-WGS hydrogen production process.

3.1 Dry Reforming of Methane (DRM)

The primary reforming reaction (Reaction 1) is strongly endothermic at +247 kJ/mol. Unlike steam methane reforming, DRM uses CO₂ as the oxidant instead of H₂O, which makes it particularly suited for biogas where CO₂ is already present in the feed in significant quantities. The theoretical H₂:CO molar ratio from DRM is 1.0, which is lower than the 3.0 from SMR. This has downstream implications for hydrogen yield, which is why WGS is needed to shift the equilibrium toward more H₂.

The Gibbs minimization approach in Aspen Plus captures several competing side reactions simultaneously. Methane cracking (Reaction 2) produces solid carbon and H₂ and is thermodynamically favorable above 600 °C. The Boudouard reaction (Reaction 3) produces carbon by disproportionation of CO and is favorable below approximately 700 °C. At 909 °C and 16 bar, the simulation predicts a solid carbon output of 0.412 kmol/hr (4.95 kg/hr), which amounts to 12.8% of the carbon in the feed appearing as coke rather than syngas products.

Coke formation is one of the recognized challenges of DRM at the industrial scale. Catalysts with high coke resistance — typically Ni-based with promoters such as Mg, Ce, or La, or noble metals

like Rh and Pt — are required for stable operation. This simulation does not model catalyst deactivation, but the carbon balance from the RGibbs output makes the coking propensity explicit. In a real system, reactor design would need to account for coke removal or suppression, for instance by adjusting the steam-to-carbon ratio or operating at slightly higher temperatures where the Boudouard equilibrium becomes less favorable.

3.2 Water-Gas Shift (WGS)

The WGS reaction (Reactions 4 and 5) is mildly exothermic at approximately -41 kJ/mol. Thermodynamic equilibrium favors higher CO conversion at lower temperatures, but kinetics are faster at higher temperatures. Industrial practice addresses this trade-off with a two-stage configuration: a high-temperature stage (typically 300 – 450 °C) over iron-chromium oxide catalyst for fast kinetics, followed by a low-temperature stage (200 – 250 °C) over copper-zinc oxide catalyst to push conversion higher.

In this simulation, both WGS stages are modeled as RStoic reactors with a fixed 75% CO conversion specification per stage. This approach simplifies the equilibrium and kinetic detail but produces results consistent with the industrial dual-stage performance. The HT-WGS inlet CO is 0.728 kmol/hr; after 75% conversion in HT-WGS and a further 75% conversion of the remaining CO in LT-WGS, the cumulative CO conversion across both stages reaches 93.75%, with a residual CO of 0.045 kmol/hr in the LT-WGS exit stream.

Adding steam before the WGS section is necessary to drive the equilibrium toward products. The steam-to-CO ratio at the HT-WGS inlet (based on stream HT-WGS-F composition) is $0.687/0.728 = 0.94$. This is on the lower end of typical industrial values (1.5 – 3.0) and reflects the limited water feed in this simulation. A sensitivity study varying steam flow rate would be a logical extension of this work.

3.3 Combustion

PSA tail gas at 1 bar contains 1.443 kmol/hr H_2 along with CO_2 , CH_4 , and CO , plus inert N_2 . Mixed with air, this provides the combustion energy to sustain the reformer's endothermic heat demand. The RGibbs combustor at 1 bar, zero heat duty, and adiabatic conditions calculates an equilibrium temperature of 2302.6 °C, with a vapor-phase product composition of 78.1 mol% H_2 , 32.0 mol% H_2O , 60.2 mol% N_2 , and essentially zero O_2 at equilibrium. All combustible components are consumed. The hot combustion gas then serves as the process heat source through the E2 and E1 heat exchangers before being vented as FLUE-GAS at 200 °C.

4. Simulation Methodology

4.1 Software and Thermodynamic Framework

All calculations were performed in Aspen Plus V14.0 (Build 359, WIN-X64 platform) under a steady-state, pressure-driven flowsheet configuration. The thermodynamic property method selected for the entire flowsheet is PR-MHV2 — the Peng-Robinson equation of state with Modified Huron-Vidal mixing rules. This choice reflects the nature of the component system: a mixture of permanent gases (H_2 , N_2 , CO , O_2), condensable species (H_2O , CH_4 , CO_2), and a solid phase (carbon graphite) at conditions ranging from ambient to 2303 °C and 1 to 16 bar. PR-MHV2 handles both the high-temperature vapor-phase equilibria in the reformer and combustor and the liquid-vapor transition in the steam generation section without switching property methods, which preserves thermodynamic consistency across the flowsheet.

Components modeled are CH_4 , CO_2 , CO , H_2 , H_2O , N_2 , O_2 , and solid carbon graphite. No other species were included. This is a simplification — trace sulfur compounds present in real biogas are omitted — but is appropriate for a process simulation focused on the hydrogen production pathway rather than feed pretreatment.

4.2 Reactor Models

Two distinct reactor model types are used, selected based on the degree of mechanistic knowledge available for each unit operation.

The reformer (REFORMER) and combustor (COMBUSTO) are both modeled as RGibbs blocks. RGibbs minimizes the total Gibbs free energy of the system subject to elemental mass balance constraints, producing the equilibrium product distribution without requiring specified reaction stoichiometry or kinetic parameters. This is appropriate for the reformer because dry reforming involves multiple simultaneous equilibria — the DRM reaction itself, methane cracking, the Boudouard reaction, and the reverse water-gas shift — whose relative extents are strongly temperature- and pressure-dependent. For the combustor, RGibbs correctly predicts complete combustion with the extremely high adiabatic temperature (2303 °C) reflecting full oxidation of the H₂-rich tail gas.

The WGS reactors (HT-WGS, LT-WGS) are modeled as RStoic blocks with a single reaction stoichiometry ($\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$) and a fixed fractional conversion of 0.75 for CO. RStoic was chosen over REquil here because the conversion specification directly encodes the industrial design basis — the 75% per-stage conversion is a known design target for the dual-stage iron/copper catalyst system — rather than relying on thermodynamic equilibrium alone. Both WGS reactors operate with zero specified heat duty (PQ flash), so the outlet temperature is a simulation output reflecting the exothermic heat release.

The PSA unit is modeled as a Sep block with defined component split fractions. This is a shortcut model: it does not simulate the adsorption cycle, but it correctly partitions the stream into a hydrogen-rich product and an impurity-rich tail gas according to the specified separation efficiency. A 79% hydrogen recovery to the tail gas (stream 19) and 21% to the product (stream H2) was specified, consistent with the simulation design basis provided.

4.3 Heat Exchanger Specifications

All heat exchangers (E1, E2, E6) modeled as HeatX blocks use countercurrent flow with a uniform overall heat transfer coefficient of 0.0203 cal/s·cm²·K. This value is a standard approximation for gas-gas heat exchange and is applied consistently to avoid introducing bias across exchangers. No pressure drop is specified on either side of any exchanger. The cold outlet temperature is the specified variable in each HeatX block: 909 °C for E1 (biogas to reformer), 350 °C for E2 (steam generation), and 250 °C for E6 (combustion air preheat). Aspen calculates the required area from the specified duty and LMTD.

Heater blocks E3, E4, E5, and E7 are simple TP-flash units with specified outlet temperature and pressure. They represent coolers in the process and do not model the heat sink explicitly — the cooling duty they report is available heat that could in principle be recovered, though this simulation does not route those duties to utility streams.

4.4 Convergence

The flowsheet contains one recycle: the combustor outlet (COMP-F) feeds back through E2 and E1 before splitting to the reformer and continuing through the process. This creates a tear stream at COMB-F. Aspen identified COMB-F as the tear stream and solved the recycle loop using the Wegstein acceleration method with a tolerance of 1.0×10^{-4} on mole fractions (with a trace fraction floor of 1.0×10^{-6} to prevent false divergence on near-zero components).

Convergence was achieved in three iterations. The final maximum relative error across all tear variables was 9.10×10^{-11} on the H₂ mole flow in COMB-F — well within tolerance. Total mole flow of COMB-F at convergence is 4.205 kmol/hr, with H₂ at 1.443 kmol/hr, N₂ at 2.182 kmol/hr,

and O₂ at 0.580 kmol/hr. All unit operation blocks and stream flashes completed normally with no warnings.

Parameter	Value	Parameter	Value
Method	Wegstein	Tolerance	1.0×10 ⁻⁴
Iterations	3	Max. Rel. Error	9.10×10 ⁻¹¹
Tear stream	COMB-F	Controlling variable	H ₂ mole flow
MAXIT	30	Status	CONVERGED

Table 2: Convergence summary for tear stream COMB-F

5. Stream Tables

Tables 3 through 6 present the complete stream data extracted from the Aspen Plus simulation. Molar flow rates are in kmol/hr, mass flows in kg/hr, temperatures in °C, and pressures in bar. Streams are grouped by process section for readability.

Stream	BIOGAS	WATER	3	4	AIR	STEAM	PEF-F
Source	Feed	Feed	C1	P1	Feed	E2	E1
Destination	C1	P1	E1	E2	B16	B9	REFORMER
Phase	Vapor	Liquid	Vapor	Liquid	Vapor	Vapor	Vapor
T (°C)	25.0	25.0	155.0	25.2	25.0	350.0	909.0
P (bar)	1.00	1.00	16.00	16.00	1.00	16.00	16.00
CH ₄ (kmol/hr)	0.8429	0	0.8429	0	0	0	0.8429
CO ₂ (kmol/hr)	0.5656	0	0.5656	0	0	0	0.5656
H ₂ (kmol/hr)	0	0	0	0	0	0	0
H ₂ O (kmol/hr)	0	0.4829	0	0.4829	0	0.4829	0
N ₂ (kmol/hr)	0.0028	0	0.0028	0	2.1818	0	0.0028
O ₂ (kmol/hr)	0.0006	0	0.0006	0	0.5800	0	0.0006
Total (kmol/hr)	1.4118	0.4829	1.4118	0.4829	2.7618	0.4829	1.4118
Total (kg/hr)	38.51	8.70	38.51	8.70	79.68	8.70	38.51

Table 3: Feed and compression section streams

Stream	8	9	HT-WGS-F	HT-WGS-P	LT-WGS-F	13	CARBON
Source	REFORMER	E3	B9	HT-WGS	E4	LT-WGS	REFORMER
Phase	Vapor	Vapor	Vapor	Vapor	Vapor	Vapor	Solid
T (°C)	909.0	350.0	350.2	551.2	210.0	264.0	909.0
P (bar)	16.00	16.00	16.00	16.00	16.00	15.70	16.00
CH ₄ (kmol/hr)	0.1686	0.1686	0.1686	0.1686	0.1686	0.1686	0
CO ₂ (kmol/hr)	0.1001	0.1001	0.1001	0.6460	0.6460	0.7825	0
CO (kmol/hr)	0.7279	0.7279	0.7279	0.1820	0.1820	0.0455	0
H ₂ (kmol/hr)	1.1444	1.1444	1.1444	1.6903	1.6903	1.8268	0
H ₂ O (kmol/hr)	0.2042	0.2042	0.6871	0.1412	0.1412	0.0047	0
C solid (kmol/hr)	0	0	0	0	0	0	0.4119
Total (kmol/hr)	2.3480	2.3480	2.8309	2.8309	2.8309	2.8309	0.4119
Total (kg/hr)	33.56	33.56	42.26	42.26	42.26	42.26	4.95

Table 4: Reformer outlet and WGS section streams

Stream	14	H2 (product)	19	16	21	FLUE-GAS
Source	E5	PSA	PSA	VALVE	B16	E7
Phase	Vapor	Vapor	Vapor	Vapor	Vapor	Vapor
T (°C)	38.0	38.0	38.0	38.0	29.4	200.0
P (bar)	15.70	15.70	15.70	1.00	1.00	1.00
CH ₄ (kmol/hr)	0.1686	0.1686	0	0	0	0
CO ₂ (kmol/hr)	0.7825	0.7825	0	0	0	0
CO (kmol/hr)	0.0455	0.0455	0	0	0	0
H ₂ (kmol/hr)	1.8268	0.3836	1.4432	1.4432	1.4432	0.2832
H ₂ O (kmol/hr)	0.0047	0.0047	0	0	0	1.1600
N ₂ (kmol/hr)	0.0028	0.0028	0	0	2.1818	2.1818
Total (kmol/hr)	2.8309	1.3877	1.4432	1.4432	4.2050	3.6250
Total (kg/hr)	42.26	39.35	2.91	2.91	82.59	82.59

Table 5: PSA, product, and tail gas streams

Stream	COMB-F	COMP-F	25	26	27	FLUE-GAS
Source	E6	COMBUSTO	E2	E1	E6	E7
T (°C)	250.0	2302.6	2128.2	1713.3	1526.3	200.0
P (bar)	1.00	1.00	1.00	1.00	1.00	1.00
H ₂ (kmol/hr)	1.4432	0.2832	0.2832	0.2832	0.2832	0.2832
H ₂ O (kmol/hr)	0	1.1600	1.1600	1.1600	1.1600	1.1600
N ₂ (kmol/hr)	2.1818	2.1818	2.1818	2.1818	2.1818	2.1818
O ₂ (kmol/hr)	0.5800	≈0	≈0	≈0	≈0	≈0
Total (kmol/hr)	4.2050	3.6250	3.6250	3.6250	3.6250	3.6250
Total (kg/hr)	82.59	82.59	82.59	82.59	82.59	82.59

Table 6: Combustion train streams

6. Unit Operation Results

6.1 Compressor C1

The three-stage centrifugal compressor raises biogas pressure from 1 bar to 16 bar with equal pressure ratios of 2.52 per stage. Stage outlet temperatures before cooling are 113.7 °C, 154.5 °C, and 155.0 °C for stages 1, 2, and 3 respectively. Inter-stage coolers in stages 1 and 2 return the gas to 60 °C. Stage 3 has no cooler — the gas exits at 155 °C directly into the heat exchange train.

Total shaft work across all three stages is 4.259 kW, with individual brake powers of 1.326, 1.472, and 1.461 kW. The isentropic efficiency is 75% for all stages. Total inter-stage cooling duty is 558.2 cal/s (198.2 + 360.1 cal/s). The pump P1 for the water feed requires only 0.0057 kW, negligible compared to the compressor.

Stage	P _{o,t} (bar)	T _{o,t} (°C, before cooling)	T _{o,t} (°C, after cooling)	Brake Power (kW)
1	2.52	113.7	60.0	1.326
2	6.35	154.5	60.0	1.472
3	16.00	155.0	155.0 (no cooling)	1.461
Total				4.259

Table 7: Compressor C1 performance summary

6.2 Reformer (REFORMER)

The RGibbs reformer at 909 °C and 16 bar converts 80% of the CH₄ feed, consistent with the product specification of 20% CH₄ remaining in the vapor phase. The heat duty required is 7534 cal/s, which is the net endothermic demand at the specified temperature. This heat is supplied indirectly via the combustion train — there is no explicit heat stream connection in the Aspen simulation, which is a modeling simplification; in practice, the reformer tube bundle sits inside the combustion furnace.

The solid carbon output of 0.412 kmol/hr is a direct simulation output from the Gibbs minimization. At 909 °C and 16 bar, the Boudouard reaction ($2\text{CO} \rightarrow \text{C} + \text{CO}_2$) is thermodynamically less favorable than at lower temperatures, but carbon is still predicted at equilibrium, likely reflecting the methane cracking pathway. This output represents 12.8% of the total carbon in the feed reporting as solid coke rather than gas-phase products, which is a significant fraction. The carbon stream exits the system and is not recycled in this simulation.

Parameter	Value
Operating temperature (°C)	909
Operating pressure (bar)	16
CH ₄ conversion (%)	80
Heat duty (cal/s)	7534 (endothermic)
H ₂ :CO molar ratio (stream 8)	1.57
Solid carbon produced (kmol/hr)	0.412
Solid carbon produced (kg/hr)	4.95
Vapor fraction at outlet	1.000

Table 8: Reformer performance summary

6.3 Water-Gas Shift Reactors

Both WGS reactors operate adiabatically. The HT-WGS outlet temperature of 551.2 °C reflects 0.546 kmol/hr CO conversion (reaction extent 0.546 kmol/hr) with the associated exothermic heat release of approximately 22.3 kJ/mol CO converted. The reaction extent in the LT-WGS is 0.136 kmol/hr, with an outlet temperature of 264 °C. Cumulative CO conversion across both stages is 93.75%, from an inlet of 0.728 kmol/hr to a residual 0.045 kmol/hr.

The H₂ content in the gas stream grows from 1.144 kmol/hr at the reformer outlet to 1.827 kmol/hr at the LT-WGS outlet — an increase of 0.683 kmol/hr attributable to the WGS reactions. This represents 59.7% more hydrogen than the reformer alone would deliver. The trade-off is 0.660 kmol/hr additional CO₂ generated in the WGS section, raising the CO₂ content from 0.100 kmol/hr to 0.782 kmol/hr in stream 13.

Parameter	HT-WGS	LT-WGS
Inlet T (°C)	350	210
Outlet T (°C)	551.2	264.0
Pressure (bar)	16.00	15.70
CO conversion (%)	75	75
Reaction extent (kmol/hr)	0.546	0.136
H ₂ produced (kmol/hr)	0.546	0.136

Parameter	HT-WGS	LT-WGS
Inlet CO (kmol/hr)	0.728	0.182
Outlet CO (kmol/hr)	0.182	0.045

Table 9: WGS reactor performance summary

6.4 Heat Exchangers

Table 10 summarizes the three HeatX units. E1 carries the largest duty at 4111 cal/s, pre-heating biogas from 155 °C to 909 °C against combustion exhaust cooling from 2128 °C to 1713 °C. The extremely high LMTD in E1 (1381.8 °C) means the required area is small despite the large duty — only 0.0147 m². The same is true for E2 (0.0043 m²) and E6 (0.0060 m²). These areas are physically small because the driving forces are enormous; in a real system, higher thermal resistance and fouling margins would push areas considerably larger.

Parameter	E1	E2	E6
Hot side in / out (°C)	2128 / 1713	2303 / 2128	1713 / 1526
Cold side in / out (°C)	155 / 909	25 / 350	29 / 250
Duty (cal/s)	4111	1757	1811
LMTD (°C)	1381.8	2027.7	1480.0
Area (m ²)	0.0147	0.0043	0.0060
UA (cal/s·K)	2.975	0.867	1.224

Table 10: Heat exchanger performance summary

6.5 PSA and Combustor

The PSA Sep block splits the cooled LT-WGS product (stream 14, 38 °C, 15.7 bar) into a hydrogen product stream and a tail gas. All CH₄, CO₂, CO, H₂O, and N₂ report entirely to the product stream H2 (alongside 21% of the H₂), while 79% of H₂ is directed to tail gas stream 19. The product stream H2 carries 0.384 kmol/hr H₂ at a purity limited by the impurities co-routed to the product side — this is a shortcut model behavior that overestimates impurity content in the product. In practice, PSA is specifically designed to exclude impurities from the hydrogen product. A more rigorous PSA model would show near-pure H₂ in the product at the expense of yield.

The combustor reaches 2302.6 °C under adiabatic equilibrium conditions. The combustion of 1.443 kmol/hr H₂ in the tail gas, along with trace CH₄, CO₂, and CO, with 0.580 kmol/hr O₂ (from 2.762 kmol/hr air) produces complete combustion — O₂ is essentially zero at equilibrium. The hot combustion gas (COMP-F) at 2303 °C then drives the steam generator E2 and biogas pre-heater E1 before being exhausted at 200 °C via E7.

The pressure reduction valve (VALVE) drops the PSA tail gas from 15.7 bar to 1 bar — a drop of 14.7 bar — to match combustor operating pressure. This is an isenthalpic expansion in the simulation, with no temperature change modeled beyond what the two-phase flash calculates (outlet temperature remains at 37.9 °C, essentially isothermal due to the near-ideal gas behavior of the stream at low flow rates).

7. Overall Mass and Molar Balance

The overall flowsheet balance is drawn across all feed streams entering the system and all product streams leaving it. Feeds are biogas (38.51 kg/hr), water (8.70 kg/hr), and air (79.68 kg/hr), giving

a total mass input of 126.89 kg/hr. Products and waste streams are: hydrogen product H₂ (39.35 kg/hr), solid carbon CARBON (4.95 kg/hr), flue gas FLUE-GAS (82.59 kg/hr), and solid carbon removed from the reformer. The Aspen overall mass balance closes to within 2.09×10^{-12} relative difference, confirming numerical integrity of the simulation.

On a molar basis the picture is more revealing. Several species are generated or consumed by reaction rather than simply passed through. Table 11 shows the overall molar balance extracted from the Aspen flowsheet balance, including the generation terms.

Component	In (kmol/hr)	Out (kmol/hr)	Net Generation (kmol/hr)
CH ₄	0.8429	0.1686	−0.674 (consumed)
CO ₂	0.5656	0.7825	+0.217 (generated)
CO	0	0.0455	+0.046 (generated)
H ₂	0	0.6668	+0.667 (generated)
H ₂ O	0.4829	1.1647	+0.682 (generated)
N ₂	2.1847	2.1847	0 (inert)
O ₂	0.5806	≈0	−0.581 (consumed)
C (solid)	0	0.4119	+0.412 (generated)
Total	4.657	5.425	+0.768 (net molar generation)

Table 11: Overall molar balance across the flowsheet

Several points stand out. CH₄ is the primary carbon and hydrogen source — 80% of the methane feed is converted across the reformer, and the remaining 20% passes through unreacted into the product stream where it is routed to the PSA and ultimately to the combustor via the tail gas. O₂ is fully consumed, confirming complete combustion. The net molar generation of 0.768 kmol/hr reflects the fact that reaction increases the number of moles — both DRM (CH₄ + CO₂ → 2CO + 2H₂, net +2 mol per mol CH₄ reacted) and combustion (H₂ + ½O₂ → H₂O, net −0.5) contribute to the overall molar change.

Carbon balance: the feed contains CH₄ at 0.843 kmol/hr and CO₂ at 0.566 kmol/hr, giving 1.409 kmol/hr total carbon. Of this, 0.412 kmol/hr (29.2%) leaves as solid coke, 0.783 kmol/hr (55.6%) leaves as CO₂ in stream 14 (directed to PSA product and then combusted or vented), and 0.168 kmol/hr (11.9%) leaves as unreacted CH₄. Only 0.046 kmol/hr (3.3%) remains as CO in the final streams. The large coke fraction is a notable feature of this simulation and would be a primary design target for optimization.

7.1 Hydrogen Yield

Hydrogen production is 0.384 kmol/hr in stream H₂, corresponding to 0.774 kg/hr of hydrogen. On a mass basis, the hydrogen yield per unit mass of biogas fed is:

$$H_2 \text{ yield} = 0.774 \text{ kg } H_2 / 38.51 \text{ kg biogas} = 0.0201 \text{ kg } H_2 / \text{kg biogas}$$

The theoretical maximum hydrogen from the biogas feed alone — if all CH₄ converted to H₂ by DRM (CH₄ + CO₂ → 2CO + 2H₂) and all CO then shifted (CO + H₂O → CO₂ + H₂) — would yield $2 \times 0.843 + 0.843 = 2.529$ kmol/hr H₂ (ignoring co-fed CO₂ contribution to WGS). Actual H₂ in stream H₂ is 0.384 kmol/hr, giving an overall hydrogen recovery of 15.2% of theoretical

maximum. This low number reflects three compounding losses: incomplete CH₄ conversion (20% unreacted), significant coke formation diverting carbon away from H₂-producing pathways, 79% of produced H₂ directed to the tail gas combustor rather than the product stream, and the PSA model routing all impurities into the product alongside the small H₂ fraction.

If the PSA were reconfigured to recover 80% of H₂ to the product (standard industrial PSA performance) and reduce coke formation, the hydrogen yield would increase substantially. This is discussed further in Section 9.

8. Energy Integration and Heat Balance

8.1 Heat Duty Summary

Table 12 compiles all thermal duties across the flowsheet, distinguishing between heat sources (positive duty, energy released) and heat sinks (negative duty, energy absorbed). The sign convention follows Aspen: heat duty is positive when heat is added to the process stream, negative when removed.

Block	Description	Duty (cal/s)	Role
REFORMER	Dry reformer (RGibbs)	+7534	Heat sink (endothermic)
COMBUSTO	Tail gas combustor (RGibbs)	0 (adiabatic)	Heat source — raises T to 2303 °C
E1	Biogas pre-heater (HeatX)	+4111	Heat recovery (hot flue → biogas)
E2	Steam generator (HeatX)	+1757	Heat recovery (hot flue → water)
E3	Syngas cooler (Heater)	−3085	Heat removal (cooling syngas to 350 °C)
E4	HT-WGS outlet cooler (Heater)	−2304	Heat removal (551 °C → 210 °C)
E5	LT-WGS outlet cooler (Heater)	−1446	Heat removal (264 °C → 38 °C)
E6	Combustion air pre-heater (HeatX)	+1811	Heat recovery (hot flue → air/tail gas)
E7	Final flue gas cooler (Heater)	−11484	Heat rejection to environment
C1 (compressor)	Biogas compression	+4259 kW equiv.	Shaft work input
P1 (pump)	Water pressurisation	+0.006 kW equiv.	Shaft work input (negligible)

Table 12: Heat duty summary across all unit operations

8.2 Heat Integration Strategy

The flowsheet uses the hot combustion exhaust — which exits the adiabatic combustor at 2303 °C — as the sole process heat source. This is the correct industrial design principle for a DRM process: the endothermic reformer requires sustained high-temperature heat, and tail gas combustion at near-stoichiometric air supply can provide it. The heat cascade in this simulation flows as follows: combustor outlet (2303 °C) → steam generation E2 (−1757 cal/s, exits at 2128 °C) → biogas pre-heating E1 (−4111 cal/s, exits at 1713 °C) → air pre-heating E6 (−1811 cal/s, exits at 1526 °C) → final cooler E7 (−11484 cal/s, exits at 200 °C).

The total heat recovered through E1, E2, and E6 is $4111 + 1757 + 1811 = 7679$ cal/s. The reformer demand is 7534 cal/s. On paper, recovered heat slightly exceeds the reformer demand — but this comparison is not directly meaningful because the heat exchangers are not connected to the reformer via an explicit heat stream in the simulation. In the actual process, the reformer tubes sit inside the combustion furnace and the heat transfer is radiation-dominated rather than convective. The simulation abstracts this as: reformer runs at 909 °C with an assigned duty, combustor provides heat by raising gas temperature to 2303 °C, and the subsequent heat exchangers recover what remains.

The cooler duties for E3, E4, and E5 (total 6835 cal/s) represent heat removed from the syngas and WGS product streams that is currently rejected as waste in this simulation. Recovering even part of this — for example, by using E3 duty to pre-heat the combustion air instead of E6, or by generating low-pressure steam from E4 and E5 — would meaningfully improve the overall energy efficiency. This is a clear optimization opportunity that the current flowsheet does not exploit.

E7 rejects 11484 cal/s to bring the flue gas from 1526 °C to 200 °C before venting. This is the largest single heat rejection in the system and represents an opportunity for external heat recovery — district heating, organic Rankine cycle, or additional steam generation — not captured here.

8.3 Overall Energy Balance

The Aspen overall enthalpy balance shows an inlet enthalpy of -28097 cal/s and an outlet enthalpy of -38427 cal/s, giving a relative difference of 26.9%. This imbalance is a simulation artifact arising from the way the reformer heat duty is handled: the reformer is assigned a fixed duty of $+7534$ cal/s (energy added to the process), but the combustor operates adiabatically and the hot gas heat cascades through E1–E6. In a fully closed energy balance, these would be explicitly linked. The enthalpy gap indicates that the simulation does not close the energy loop between the combustor and the reformer with an explicit heat transfer stream — a recognized limitation of this flowsheet architecture. Despite this, all stream enthalpies and unit operation duties are individually consistent; the issue is at the global accounting level, not in the local unit operation calculations.

9. CO₂ Equivalent Analysis

Aspen Plus tracks CO₂-equivalent values for all streams using the standard CO₂ global warming potential weighting. Table 13 summarises the CO₂e flows across the process boundary.

Stream / Category	CO ₂ e (kg/hr)	Note
Feed streams (BIOGAS)	402.15	Includes CH ₄ GWP contribution (1 kg CH ₄ = 25 kg CO ₂ e)
Product streams (H ₂ , FLUE-GAS, CARBON, stream 14)	109.89	Residual CO ₂ and trace CH ₄ in product streams
Net CO ₂ e reduction	-292.26	CO ₂ e removed from atmosphere via biogas utilisation
Utilities CO ₂ e	0	No external utility streams modeled
Total net CO ₂ e production	-292.26	Net negative — process is a CO ₂ sink on GWP basis

Table 13: CO₂ equivalent summary

The large CO₂e value for the biogas feed (402.15 kg/hr) is dominated by the methane GWP contribution. Methane carries a GWP of approximately 25 over a 100-year horizon, so 0.843

kmol/hr CH₄ (13.5 kg/hr) contributes roughly 338 kg/hr CO₂e, with the actual CO₂ in the feed adding the remaining 64 kg/hr. When biogas is burned or vented without utilisation, these 402 kg/hr CO₂e are released to the atmosphere. This simulation shows that routing the same biogas through the DRM-WGS process converts most of the CH₄ and fixes the CO₂ into syngas products, reducing the net CO₂e output to 109.89 kg/hr — a 72.7% reduction in climate impact relative to direct biogas combustion or venting.

This framing is important for positioning biogas-based hydrogen in the green hydrogen taxonomy. Hydrogen produced from fossil natural gas via SMR carries a direct CO₂ burden from combustion; hydrogen from grid electrolysis carries an indirect burden from electricity generation. Biogas-DRM hydrogen, if the feedstock is genuinely waste-derived (landfill gas, sewage sludge digestate, agricultural slurry), starts from a carbon-negative accounting position because the CH₄ would otherwise reach the atmosphere. The 292.26 kg/hr net CO₂e reduction in this simulation represents the climate credit from diverting that methane into a productive hydrogen supply chain rather than releasing it.

The caveat is that this analysis covers only the direct process streams. A complete life-cycle assessment would include embodied emissions from catalyst manufacturing, compressor electricity supply, construction, and end-of-life. Those factors can shift the net balance, but the process-level CO₂e accounting presented here establishes a strong baseline case.

10. Process Performance Metrics

Table 14 consolidates the key performance indicators from the simulation into a single reference summary.

Metric	Value	Unit
Biogas feed flow	38.51	kg/hr
Water feed flow	8.70	kg/hr
Air feed flow	79.68	kg/hr
Reformer temperature	909	°C
Reformer pressure	16	bar
CH ₄ conversion (reformer)	80	%
HT-WGS CO conversion	75	%
LT-WGS CO conversion	75	%
Cumulative CO conversion (both WGS stages)	93.75	%
H ₂ produced (total in process)	1.827	kmol/hr
H ₂ in product stream H2	0.384	kmol/hr
H ₂ mass in product stream H2	0.774	kg/hr
PSA H ₂ separation efficiency (to product)	21	% (79% to tail gas)
H ₂ yield per kg biogas	0.0201	kg H ₂ / kg biogas
Solid carbon (coke) produced	4.95	kg/hr
Combustor adiabatic temperature	2302.6	°C
Total compressor work	4.259	kW

Metric	Value	Unit
Reformer heat duty	7534	cal/s
Total heat recovered (E1 + E2 + E6)	7679	cal/s
Net CO ₂ e reduction	-292.26	kg CO ₂ e / hr
Flowsheet convergence	3 iterations	Wegstein, tol. 1×10^{-4}
Max. relative convergence error	9.10×10^{-11}	Dimensionless

Table 14: Consolidated process performance summary

10.1 Comparison with Literature Benchmarks

Industrial DRM-WGS processes for hydrogen production are less documented than SMR in the open literature, but several pilot and demonstration studies provide reference points. Reported CH₄ conversions in DRM at 800–950 °C range from 70 to 95% on active Ni and noble metal catalysts, consistent with the 80% conversion set in this simulation. H₂:CO ratios from DRM typically fall between 0.8 and 1.2, and the simulated value of 1.57 at the reformer outlet is slightly above this range, likely because the RGibbs equilibrium model at 909 °C favours a slightly elevated H₂ fraction relative to kinetically limited experimental results.

PSA hydrogen recovery in commercial units typically reaches 70–85%, with product purity above 99.9%. The simulation PSA recovery to product (21%) is far below the commercial range — this is a deliberate conservative setting applied in the simulation basis, not a claim about PSA technology capability. If the PSA split were reversed (21% to tail gas, 79% to product), the hydrogen product flow would increase to approximately 1.443 kmol/hr (2.91 kg/hr), raising the H₂ yield from 0.0201 to roughly 0.0756 kg H₂/kg biogas — still modest but more representative of what the process could deliver with a standard PSA unit.

The coke production of 4.95 kg/hr from 38.51 kg/hr biogas (12.9% by mass of biogas feed) is on the higher end of what equilibrium models predict for DRM at 16 bar. Elevated pressure favours carbon formation thermodynamically (Le Chatelier: coke formation reduces gas-phase moles, which pressure suppresses for gas-expanding reactions but favours for Boudouard which produces a net reduction in gas moles at high CO partial pressures). Industrial DRM operation typically uses low pressures (1–5 bar) or catalyst modifications to suppress coke. Operating at 16 bar without a coke suppression mechanism would likely lead to rapid catalyst deactivation in practice.

11. Limitations and Assumptions

Every simulation study has a boundary beyond which the model does not speak, and being explicit about those boundaries is part of honest technical reporting. The following assumptions and limitations apply to this work.

Equilibrium reactor models (RGibbs) assume that thermodynamic equilibrium is reached at the specified temperature and pressure. In reality, reaction rates are finite and catalyst activity determines how closely the system approaches equilibrium. The reformer equilibrium assumption likely overestimates conversion at the specified temperature relative to a kinetically limited real reactor, particularly for the slower DRM reaction relative to side reactions. Similarly, the combustor equilibrium assumption is reasonable given the very high temperature, but trace species may not fully equilibrate.

The PSA is modeled as a component splitter, not a dynamic adsorption cycle. This means the simulation cannot capture the effect of PSA operating pressure, cycle time, adsorbent loading, or purge gas requirements. The split fractions were specified directly from the design basis. A more

rigorous model would use a detailed PSA cycle simulation with isotherm data for the relevant adsorbents (zeolite 5A or activated carbon for CH₄/CO₂/CO removal, and zeolite 4A or activated carbon for H₂ purification).

No catalyst deactivation or pressure drop across reactor beds is modeled. In practice, the reformer and WGS reactors would show pressure drop on the order of 0.5–2 bar per reactor bed, which affects downstream compression requirements and equilibrium conditions.

The heat balance between the combustor and the reformer is not explicitly closed — the reformer operates with an assigned duty and the combustor operates adiabatically, but no direct heat stream links them. The enthalpy discrepancy of 26.9% in the overall flowsheet balance reflects this. A more complete model would use a furnace block or a heat stream connecting the combustor outlet enthalpy to the reformer duty.

Water feed is provided as pure liquid at 25 °C and 1 bar. In a biogas plant context, process water may require demineralisation and pretreatment, adding to operating cost and energy consumption not captured here.

No economic analysis, capital cost estimation, or techno-economic assessment is included. The simulation provides the physical and thermodynamic basis for such an analysis but does not extend to it.

12. Recommendations for Future Work

The simulation as built provides a solid foundation but leaves several productive directions open.

The most immediate priority is a sensitivity analysis on the key operating variables. Reformer temperature (800–1000 °C), pressure (1–20 bar), CH₄:CO₂ feed ratio (reflecting different biogas compositions), and steam-to-carbon ratio at the WGS inlet all affect H₂ yield and coke production in ways that this single-point simulation does not capture. Aspen Plus sensitivity tools make this straightforward to implement as a next step.

The PSA specification deserves particular attention. Reversing the split to 79% H₂ recovery to the product (the realistic direction for industrial PSA) and running the simulation to check whether the tail gas still provides sufficient combustion energy for the reformer would validate the heat integration closure. If the tail gas H₂ content falls too low, a supplementary fuel source or partial air pre-heating might be needed.

Coke suppression strategies should be evaluated. Adding steam to the reformer feed (effectively moving toward a combined steam and dry reforming, or CSDRM, configuration) reduces coke formation by the reverse Boudouard and steam gasification reactions. A parametric study varying the steam-to-methane ratio at the reformer inlet would identify the operating window that keeps carbon formation below an economically tolerable threshold.

On the energy side, heat integration of the cooler duties (E3, E4, E5 total: 6835 cal/s) into the combustion air pre-heating or steam generation circuits would reduce E7 reject duty and improve overall thermal efficiency. A pinch analysis using Aspen Energy Analyzer would identify the minimum utility targets and the optimal heat exchanger network for this flowsheet.

Finally, replacing the SEP block PSA model with a rigorous cycle simulation — or coupling the Aspen flowsheet to a dedicated PSA simulator — would give more reliable product purity and recovery predictions and close the remaining gap between this simulation and a bankable process design.

13. Conclusions

This study has presented a steady-state Aspen Plus simulation of a hydrogen production process from biogas via dry reforming and dual-stage water-gas shift, with PSA purification and integrated tail gas combustion for heat supply. The simulation converged cleanly in three Wegstein iterations with a maximum relative error of 9.10×10^{-11} , confirming numerical stability of the recycle loop.

From a 38.51 kg/hr biogas feed (59.7 mol% CH₄, 40.1 mol% CO₂), the process produces 0.774 kg/hr of hydrogen in the product stream under the specified PSA configuration, with 80% CH₄ conversion in the RGibbs reformer at 909 °C and 16 bar. The dual-stage WGS section achieves a cumulative CO conversion of 93.75%, increasing the hydrogen content from 1.144 kmol/hr at the reformer outlet to 1.827 kmol/hr before the PSA. The tail gas combustor operates at 2302.6 °C adiabatically, and the heat cascade through E1, E2, and E6 recovers 7679 cal/s against a reformer demand of 7534 cal/s — a heat-integrated design that avoids any external fuel input beyond the biogas itself.

The CO₂-equivalent analysis shows a net reduction of 292.26 kg CO₂e/hr relative to venting or direct combustion of the biogas feed, placing this process in a net-negative climate accounting position when the full GWP of the methane feed is credited. This is a meaningful differentiator for biogas-derived hydrogen in a policy environment that values avoided methane emissions.

The simulation identifies three areas that limit performance in the current configuration. First, the coke production of 4.95 kg/hr (0.412 kmol/hr solid carbon) represents 29% of the feed carbon diverted away from hydrogen-producing pathways — a consequence of operating DRM at elevated pressure without steam addition or other coke suppression. Second, the PSA configuration directs 79% of produced hydrogen to the combustor tail gas rather than the product stream, severely limiting product hydrogen yield. Third, the energy balance between the combustor and the reformer is not explicitly closed in the simulation architecture, leaving a 26.9% enthalpy discrepancy that a more complete model would resolve with an explicit heat stream connection.

Taken together, the simulation demonstrates that the DRM-WGS route from biogas to hydrogen is thermodynamically consistent, convergent, and capable of net-negative CO₂e operation. The specific performance numbers reported here reflect the design basis choices made — PSA split, reformer constraint, adiabatic WGS — and the real process could deliver substantially higher hydrogen yields with optimized PSA recovery, reduced coke, and a steam co-feed. This simulation establishes the quantitative foundation from which those optimizations can be systematically evaluated.

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